Machine Learning Based Design of an Electronic Scarecrow for Protecting Agricultural Yields(Rice) against Predators

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Segu Mohammed

Outline

1. Introduction
2. The Problem
3. Project Objectives
4. System Design and Implementation
   - ELECTRICAL , SOFTWARE & MECHANICAL
5. Analysis, Testing & Results
6. Conclusion, Limitations & Future works

Introduction

Introduction

1 Economic Losses

Predators, particularly birds, pose a significant threat to agricultural production in Africa, causing substantial economic losses and jeopardizing food security for local communities.

2 Inadequate Traditional Methods

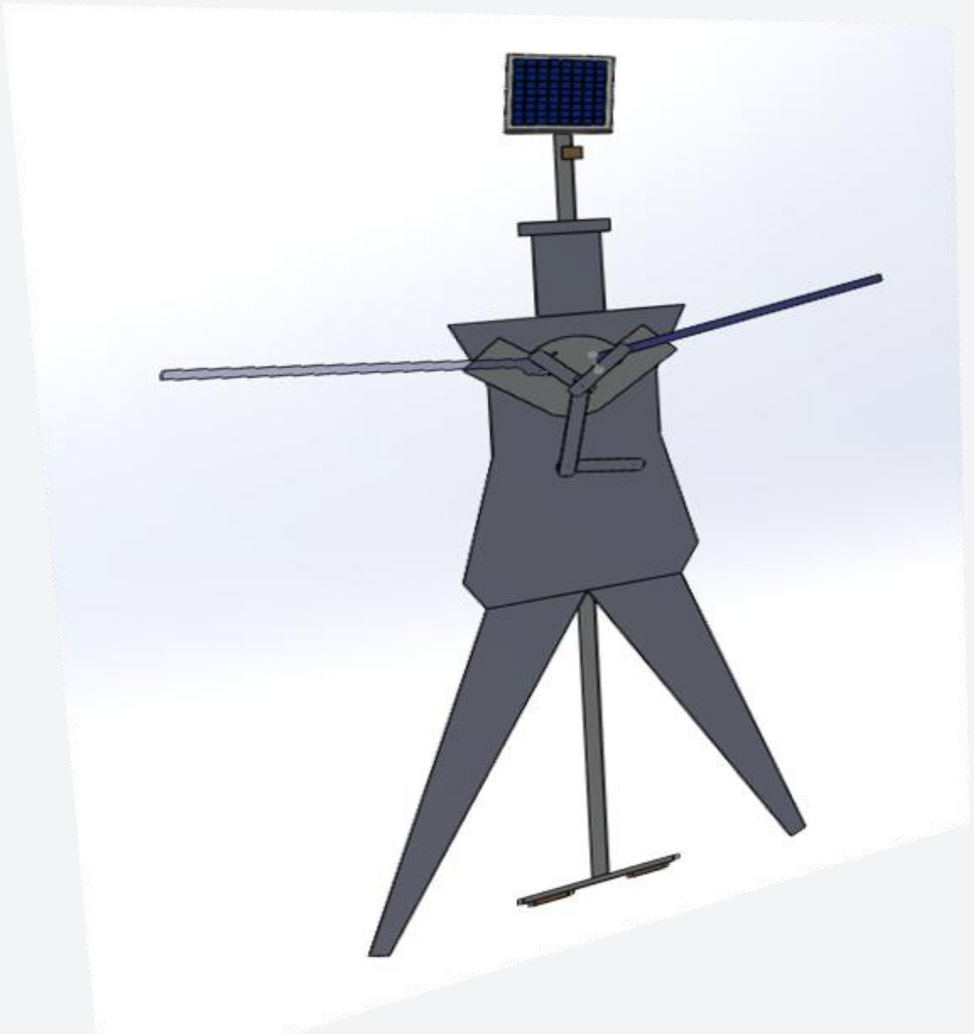
Traditional predator deterrence methods, such as scarecrows, often lack adaptability and responsiveness, relying on static stimuli and failing to adjust based on real-time predator behavior.

From research:

- Physical chasing, shouting, and scaring off
- Beating sonorous bodies like tins and jerry cans
- Poisoning and trapping
- Use of stationary scarecrows

Need for Dynamic Solution

The "Shield Your Farm" project aims to address these limitations through the development of an intelligent electronic scarecrow that integrates machine learning technologies.



Project Objectives

Machine Learning System Development

Develop and implement machine learning algorithms capable of analyzing real-time camera feed to detect potential bird on rice farms.

Responsive Bird-Scaring Mechanisms

Design and integrate responsive bird-scaring mechanisms, such as sound from speakers, two-dimensional movement mechanisms, and a traditional way of scaring birds, triggered an action when the machine learning model detect birds in the rice farm..

System Design and Implementation

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Electrical , Software & Mechanical

Design Requirements

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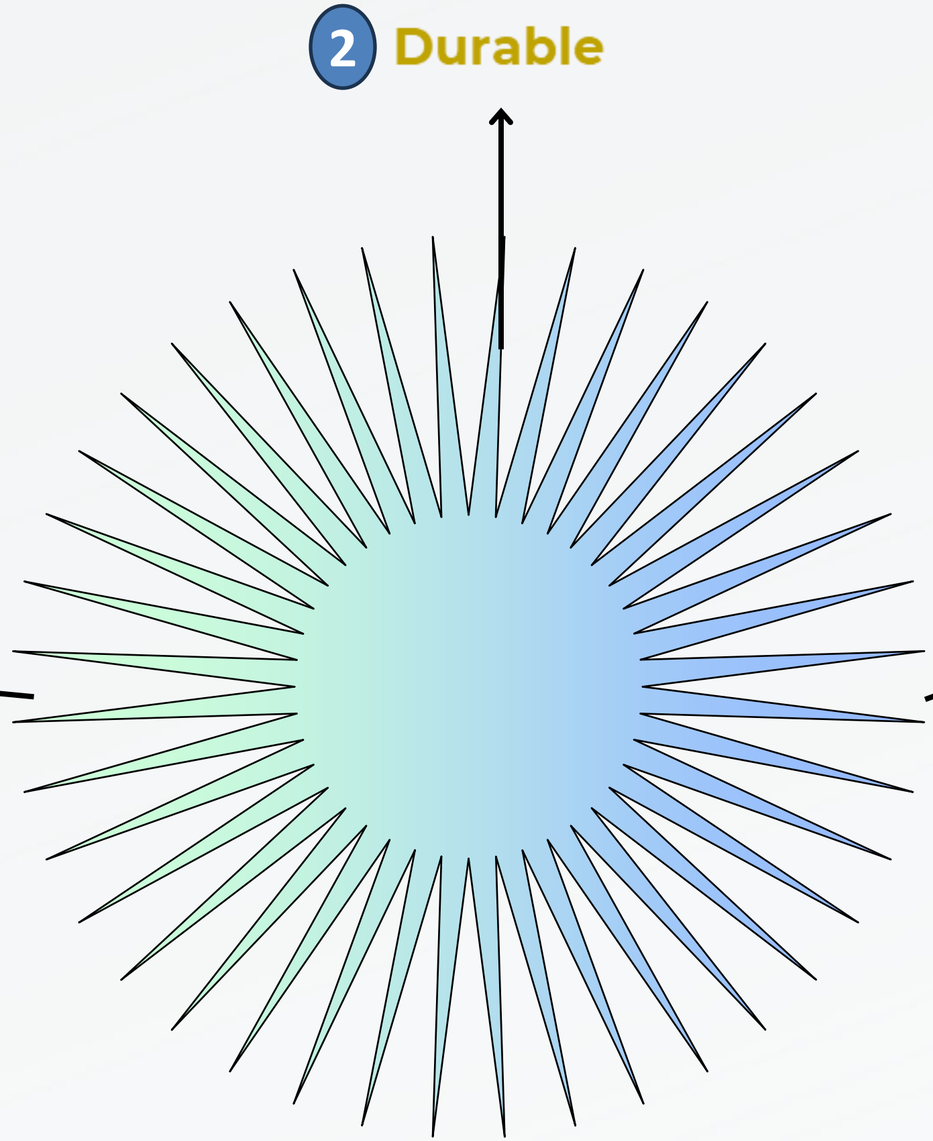
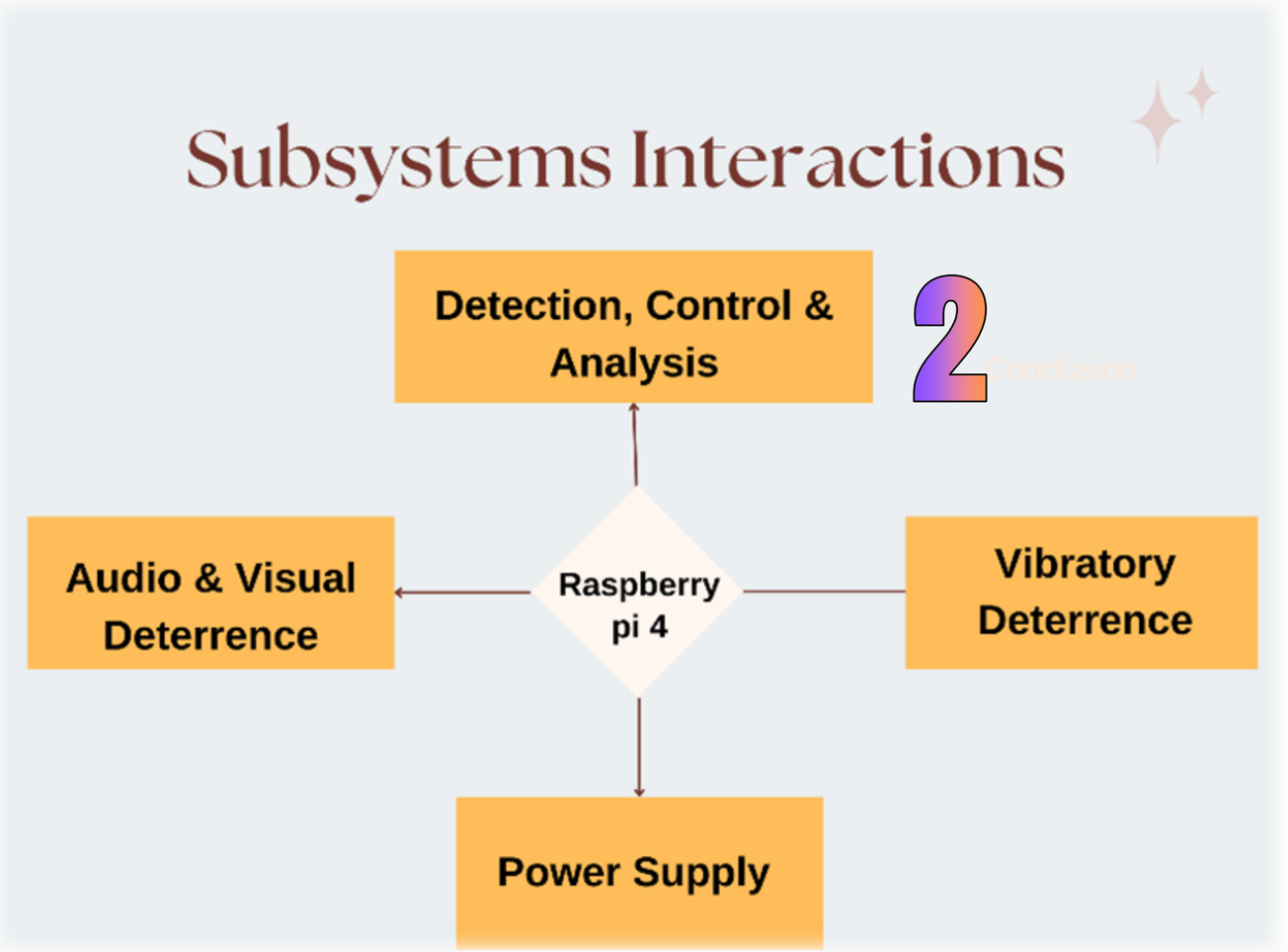


1. Efficient
2. Durable
3. Scalability

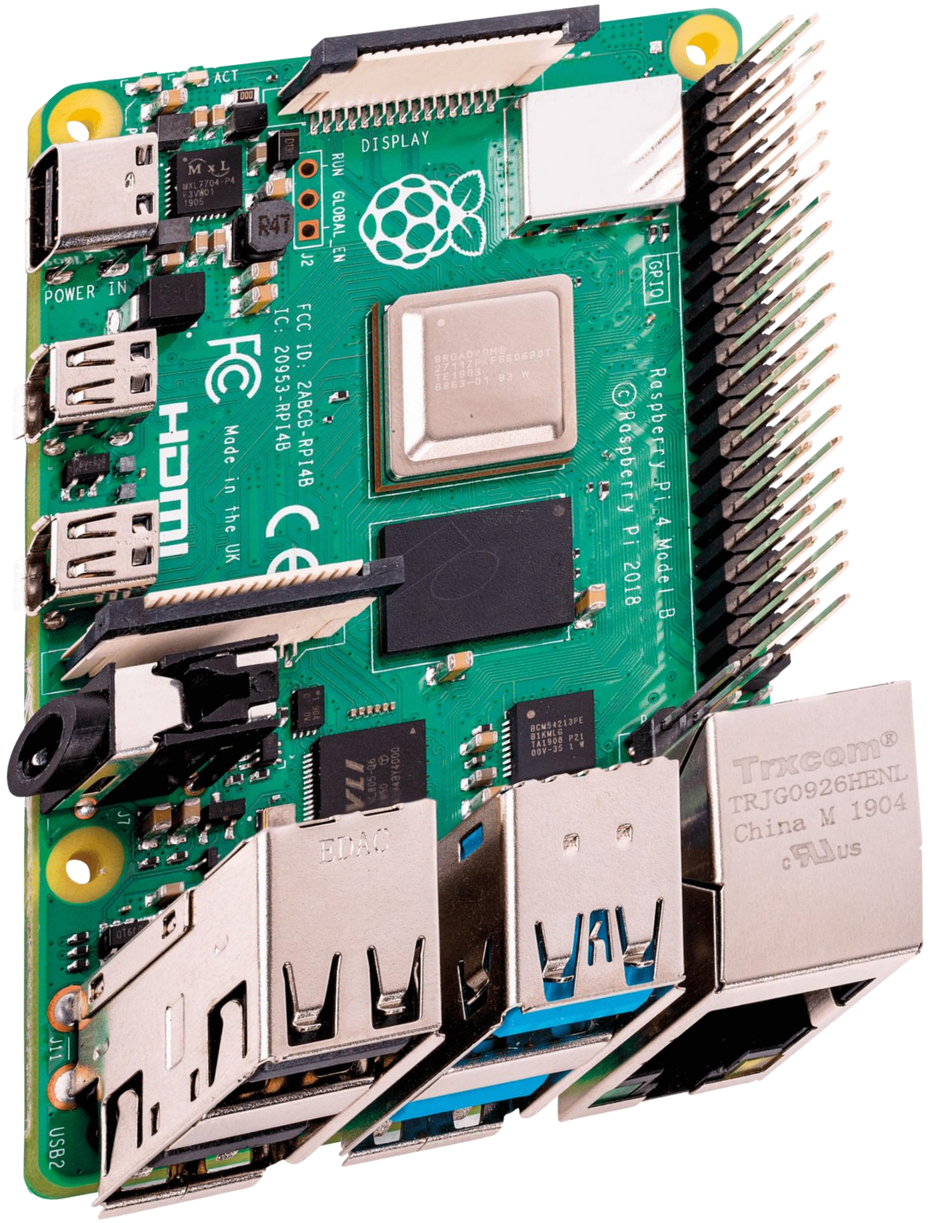
Diagram of Subsystems



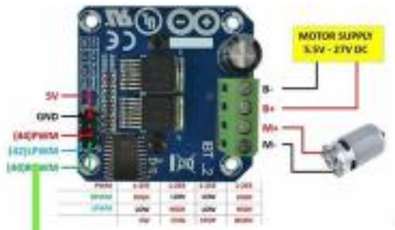
Microcontroller Selection

Criteria	Raspberry Pi 4 B (baseline)	Weight (Out of 5)	ESP32 Cam	Arduino Uno	STM32f1	Node MCU (ESP8266)
High processing power	0	3	-2	-3	-3	-2
Speed	0	2	-1	-3	-3	-2
Availability	0	1	+1	+2	-2	+1
Cost	0	1	+1	+1	-2	+2
Voltage	0	1	+1	+2	+1	+1
TOTAL	0	8	-5	-10	-18	-6

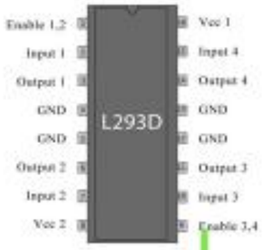
The Raspberry Pi 4 was chosen as the central MCU due to its robust processing capabilities and connectivity options, making it an excellent choice for serving as the project's backbone. Its ability to handle complex tasks and manage multiple connections lays a solid foundation for future expansion and integration of additional functionalities.



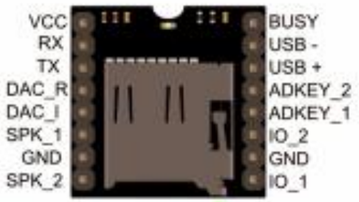
Design – Electrical Components Selection



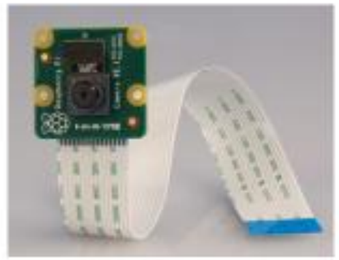
BTS 7960



L293D



DFRobot mp3



Picamera v2

775 DC

MG995R Servo

Speaker

5V buck coverter

How reliable is the power source to the system?

Components	Current Ratings (A)	Voltage Ratings (V)
MG995 Servo Motor (x2)	1.2	6
Raspberry 4 B	3.00	5.00
775 Motor	1.2	12.00
L298N Motor Driver	0.036	3.2
L293D Motor Driver	0.6	5

How reliable is the power source to the system?

$$PT = [14.4 + 15 + 14.4 + 0.1152 + 3]\, W$$

$$PT = 46.9152W$$

Battery Specifications:

- $Voltage = 12V$
- $Capacity = 17Ah$
- Power: 204 watts-hour

The Power Source

Power Source

Solar Panel

Solar
Charge
controller

Load

Electrical Schematics

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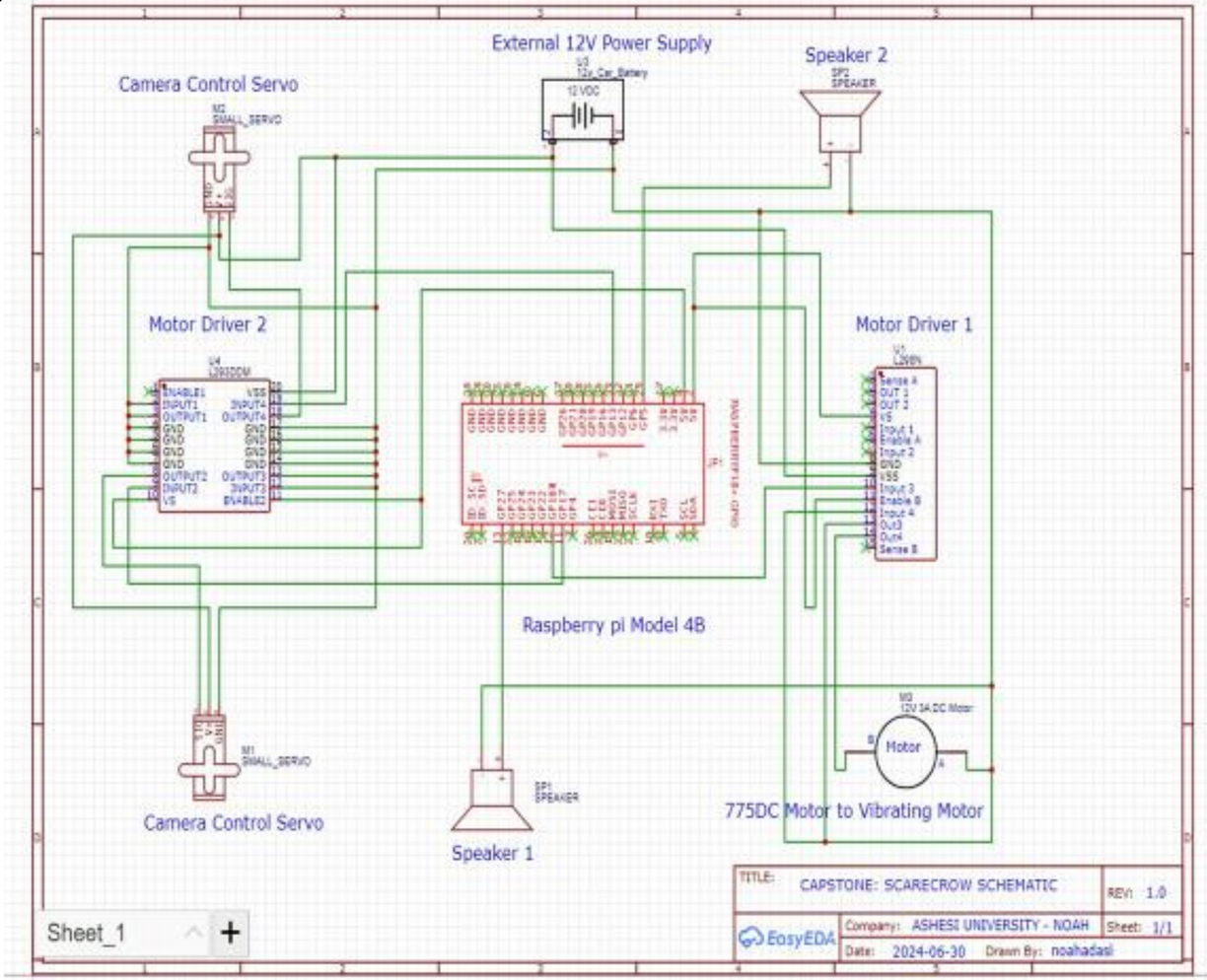
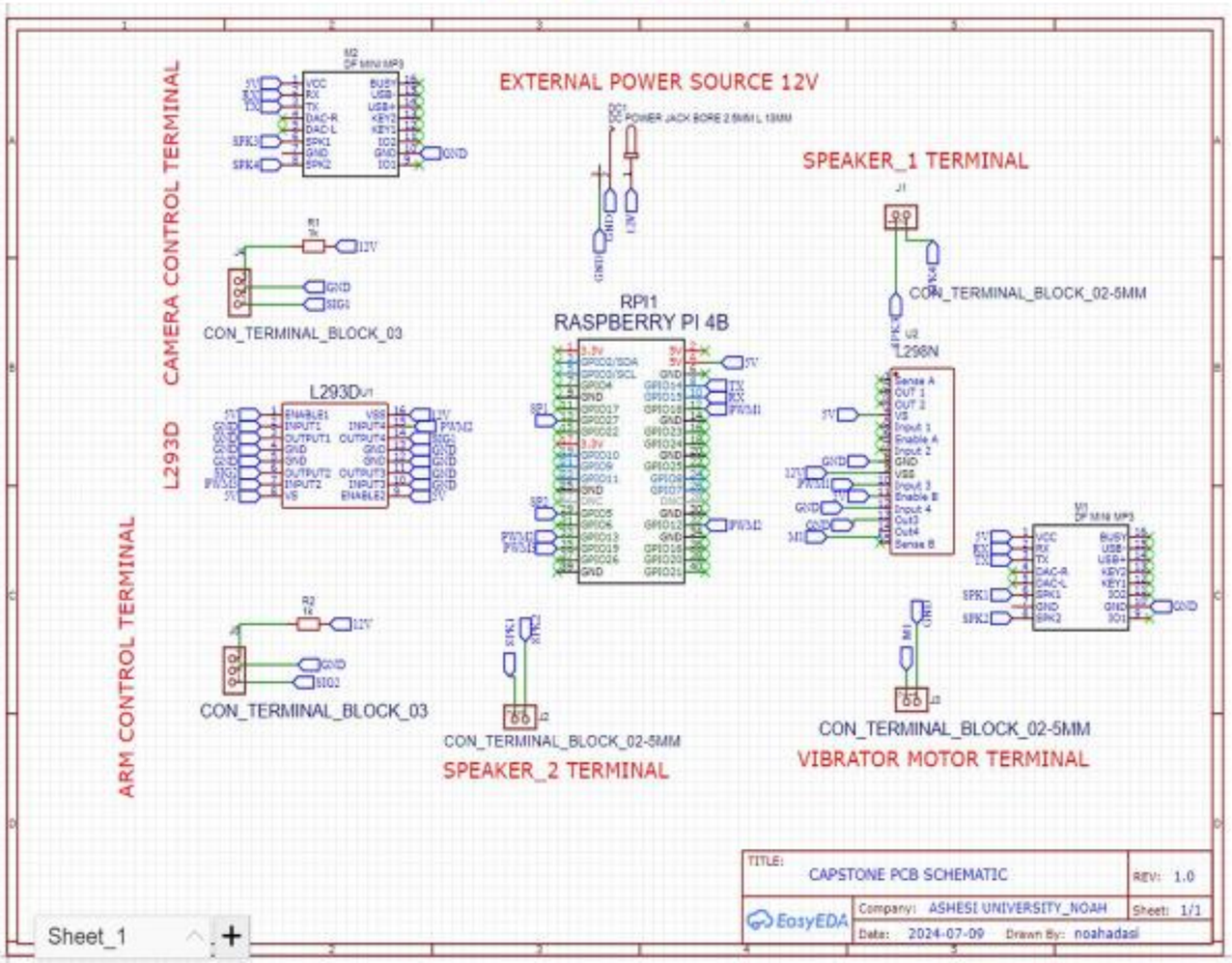
Conceptual schematic



PCB schematic



PCB Design

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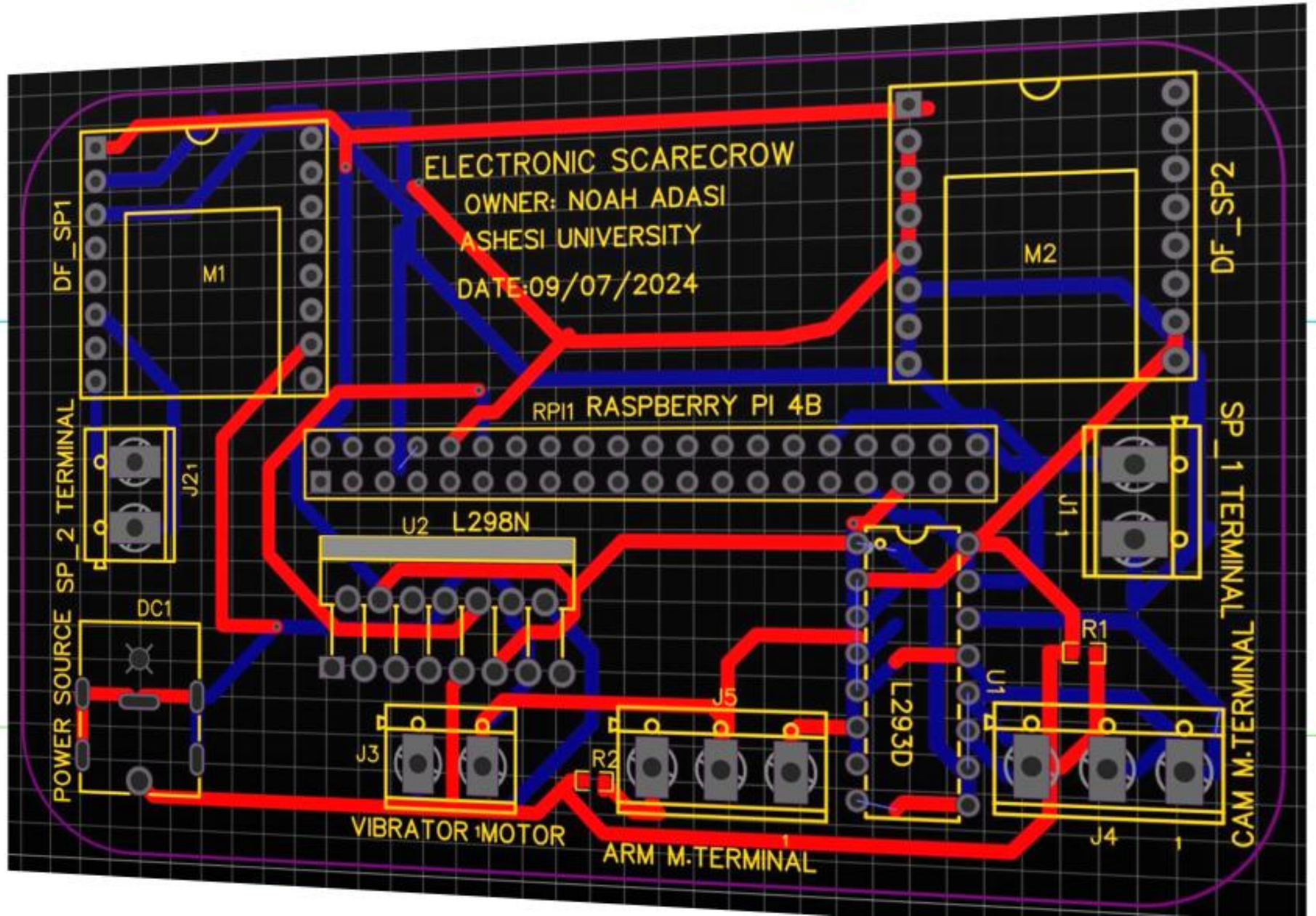

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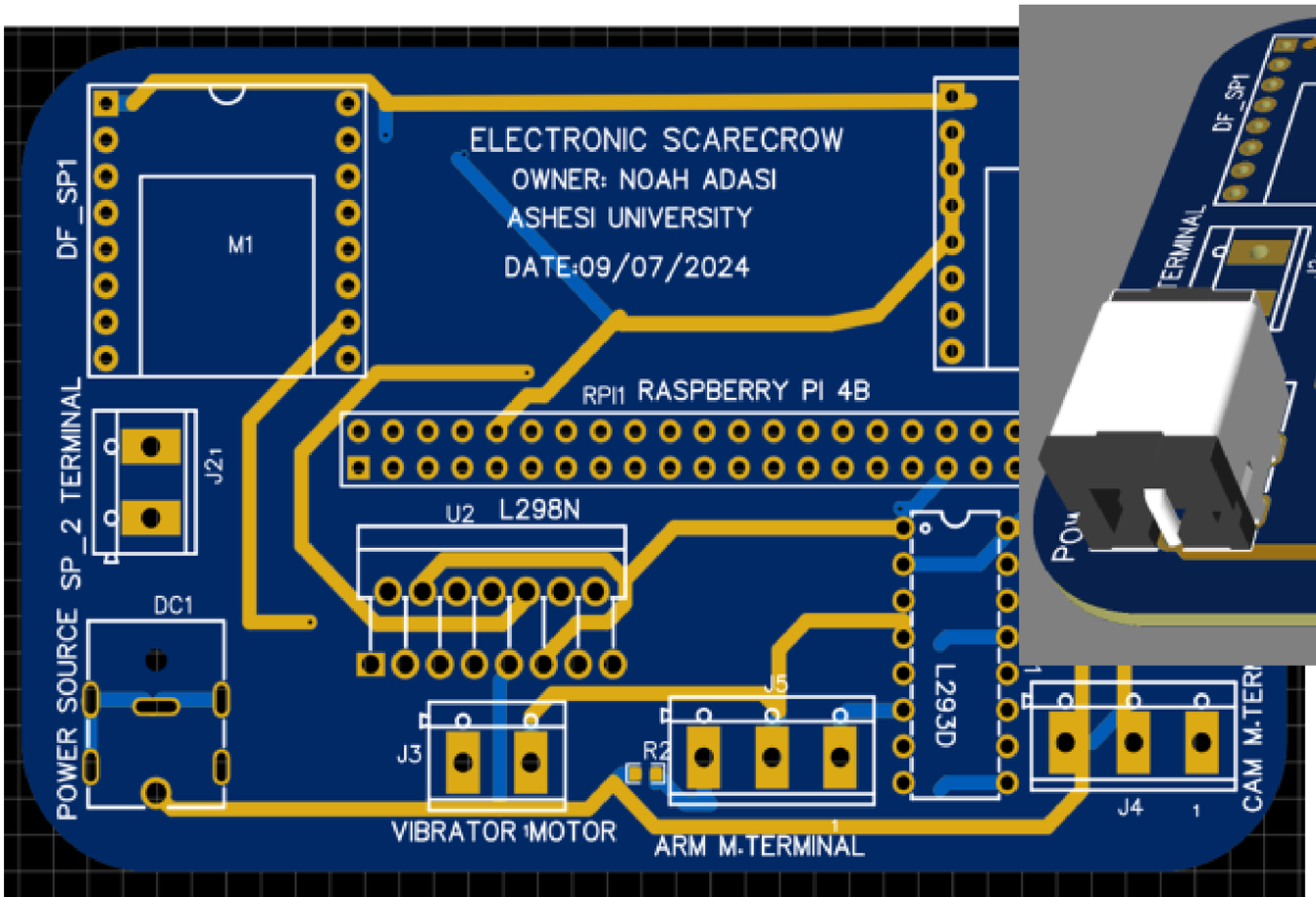
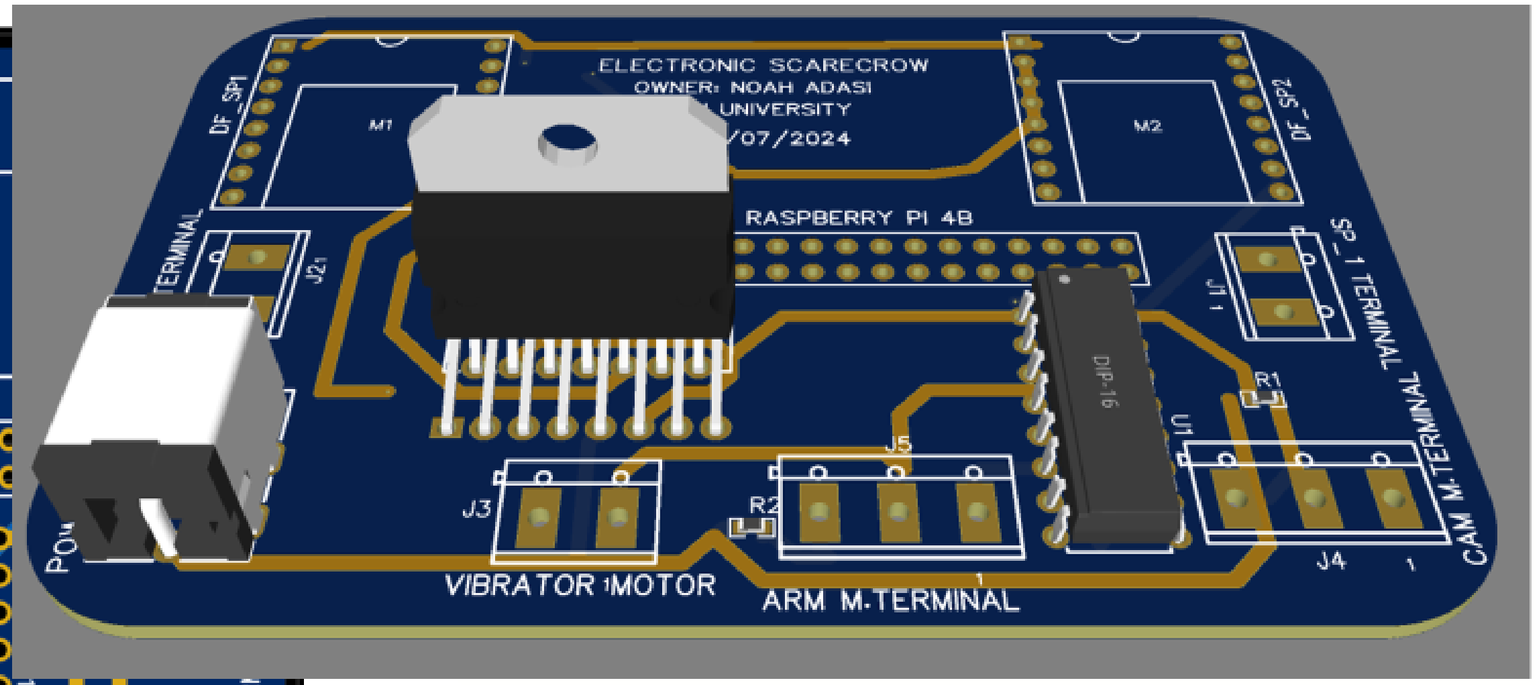
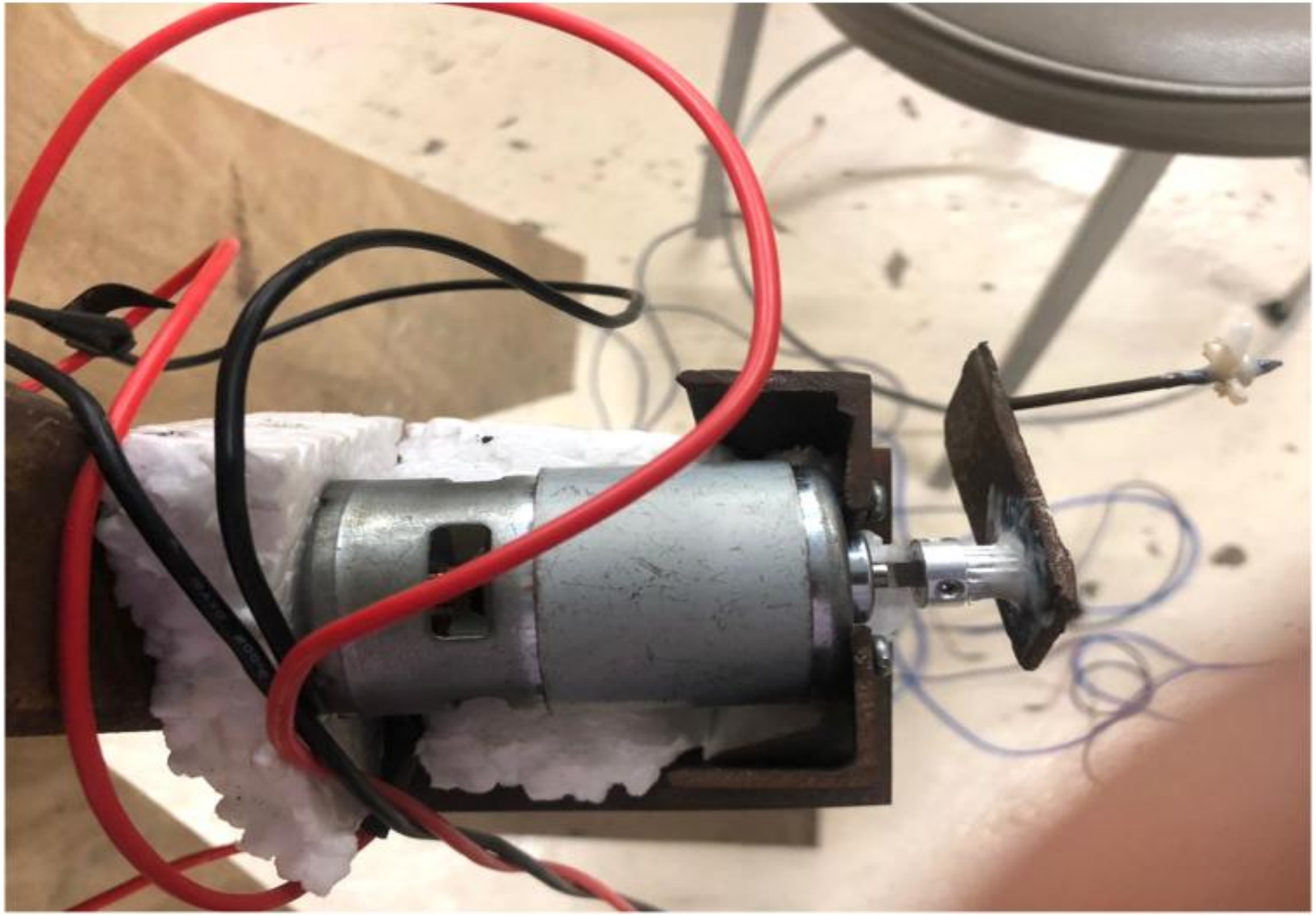
Electrical: PCB Realization 2D & 3D





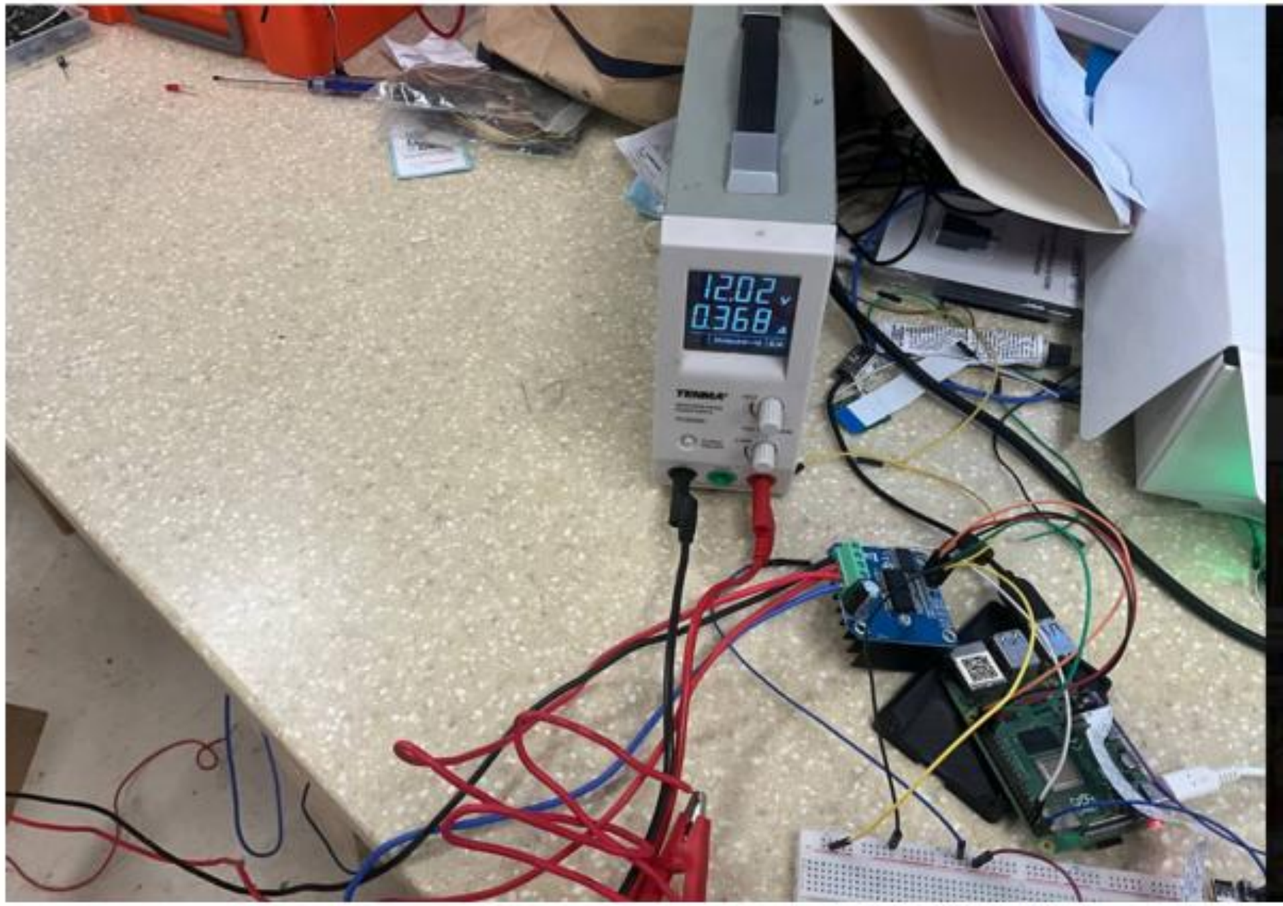


Vibrator motor

Parameters Considered

Impact of Mass on Vibration

- Vibration Intensity: A larger mass can increase the vibration intensity, but it also requires more torque to move



Vibrator motor

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Training a YOLOv8 Model for Birds Detection

Introduction to YOLOv8

1 What is YOLOv8?

YOLOv8 stands for "You Only Look Once" version 8, an advanced real-time object object detection model. It is known for its speed, accuracy, and efficiency, making it making it suitable for various applications.

2 Key Features

YOLOv8 boasts several key features, including its ability to process images in real- in real-time with minimal latency, high precision in detecting and localizing multiple localizing multiple objects, and its design for performance on both high-end GPUs end GPUs and edge devices.

3 Applications

YOLOv8 finds applications in diverse fields such as autonomous driving, surveillance, surveillance, retail, and healthcare, where real-time object detection is crucial. crucial.

Dataset Preparation

Preprocessed Dataset was used

Other details:

10000 birds' images with 100 bird's species

Single Image Dimension: 640 (height) x 640 (width) x 3 (channels) = 1,228,800 pixels

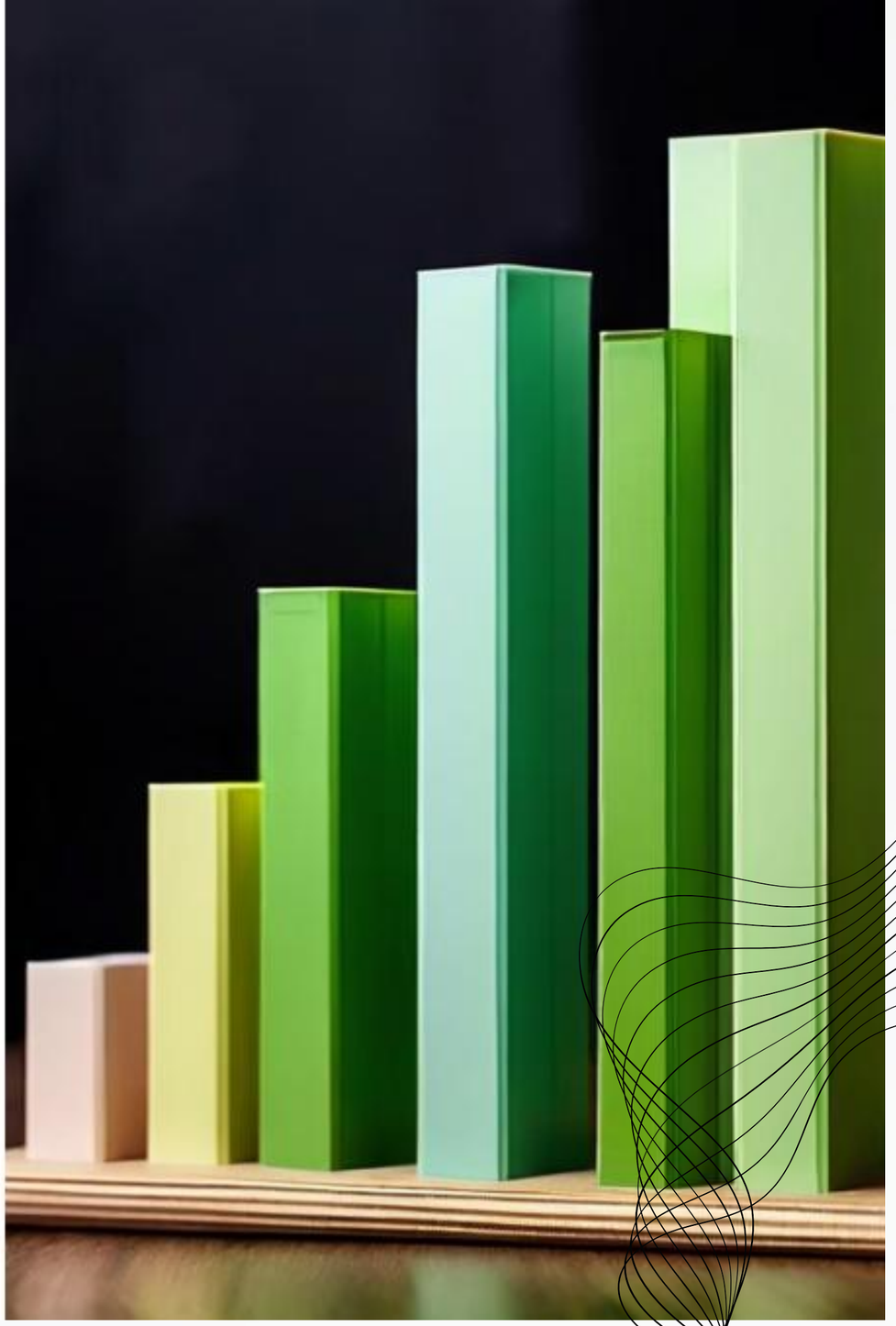
- Sourced from preprocessed publicly available datasets from Kaggle
- **Annotation Process:** LabelImg, an open-source graphical image annotation tool. Each image was annotated with a corresponding .txt file containing YOLO format annotations, including class labels and bounding box coordinates.
- All images resized to 640x640 to ensure uniformity in input size.
- Channels: 3 (RGB)

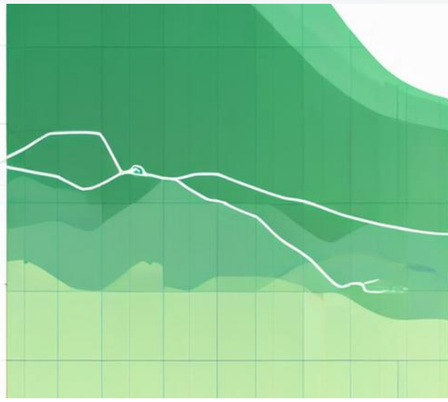
Model Evaluation

Evaluation Metrics

The yolov8's model's performance was evaluated using metrics like precision, recall, and mAP (mean Average Precision). These metrics provided insights into the model's accuracy and ability to correctly identify birds.

•Precision (98%): Out of all the birds detected by the model, 98% were correctly identified.
•Recall (94%): The model successfully detected 94% of all the actual birds present in the images.
•mAP (90%): The model achieved an average precision of 85% across all birds' classes and different levels of recall.





Training Parameters

Parameter	Value	Rationale
Epochs	100	Ensures sufficient training and prevents overfitting.
Batch Size	16	Balances computational load and convergence speed while considering memory constraints.
Image Size	640x640	Ensures compatibility with YOLOv8 architecture and maintains sufficient detail for accurate detection.

Learning rate: The initial Learning rate was set to 0.01 and then I decided to increase it by 0.1 in every 20 epoch

Hardware used & other considerations

Hardware Setup

1

The training was conducted on Google Colab, leveraging the NVIDIA Tesla T4 GPU for accelerated computations. The Colab environment provided a standard multi-core CPU and approximately 12-16GB RAM.

Runtime

3

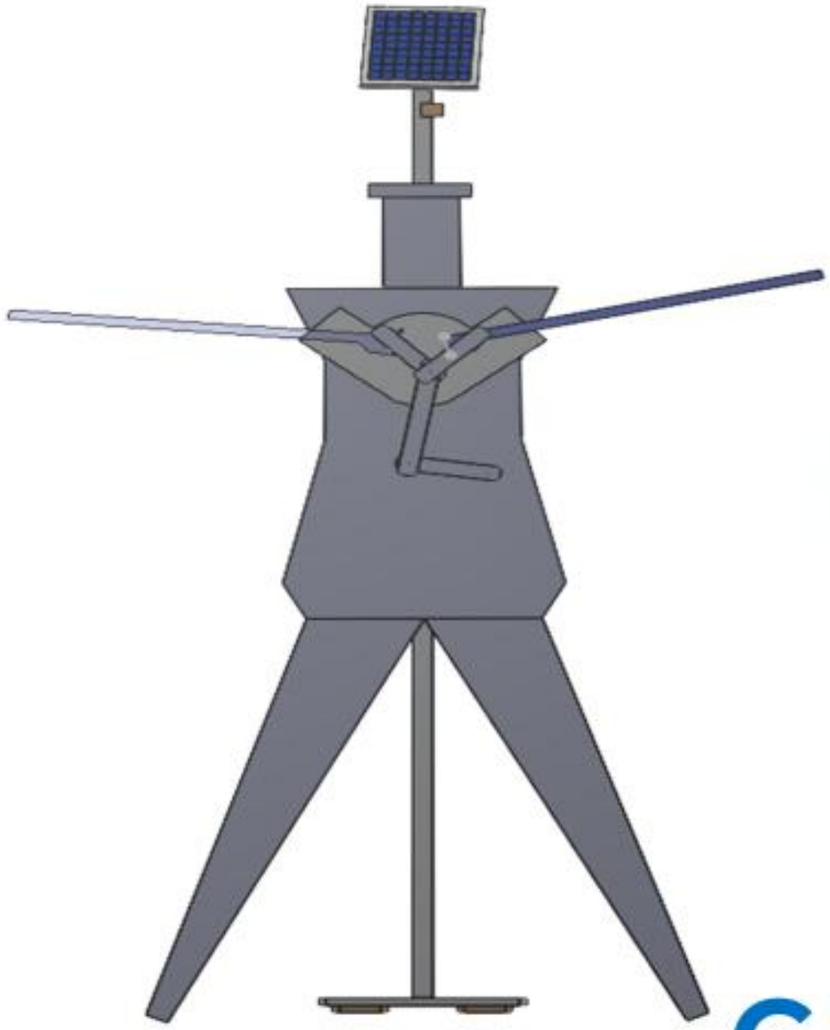
The training process took approximately 12 hours for 100 epochs using.

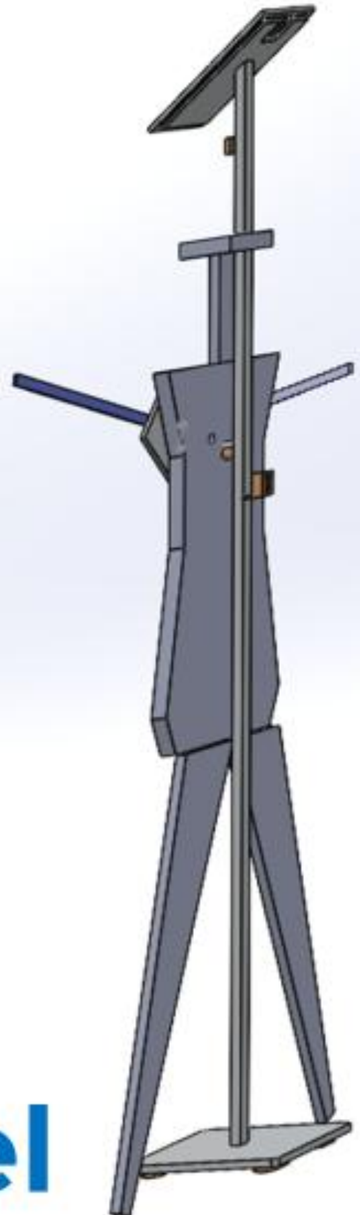
2

Training Environment

The Ultralytics YOLOv8 library with PyTorch backend was used for training. Challenges included memory management, adjusting batch size and image size to fit within GPU memory,

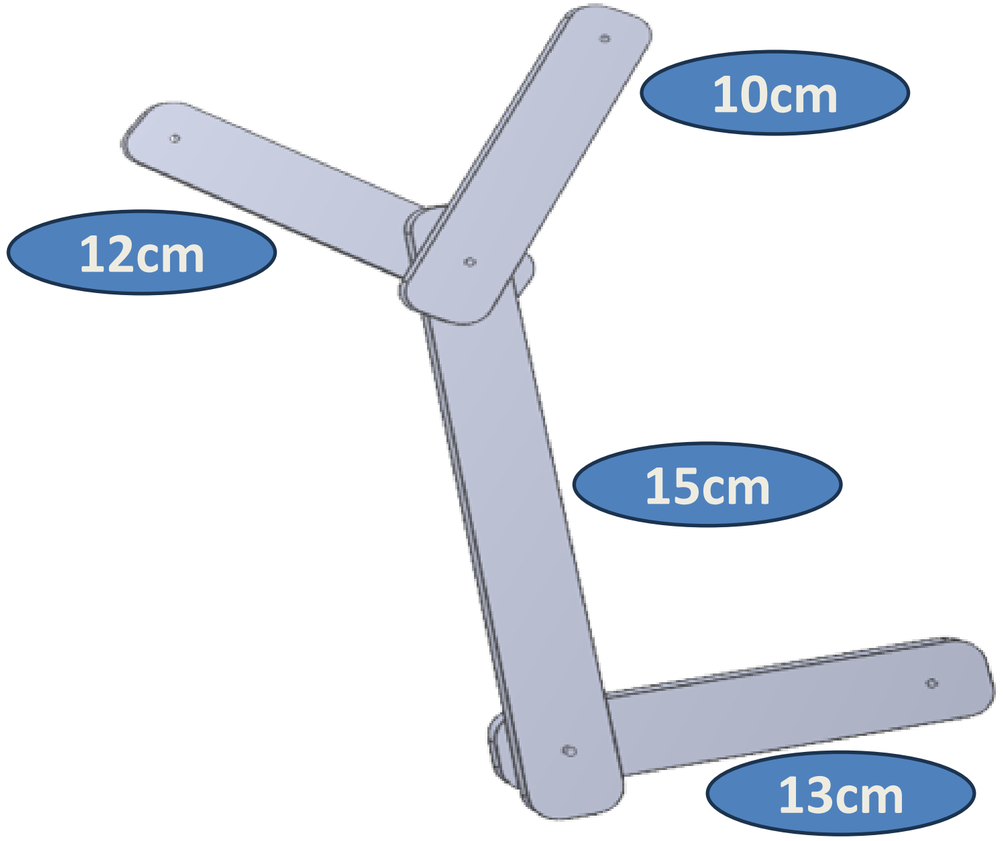
Mechanical Design





CAD Model

Mechanical Design



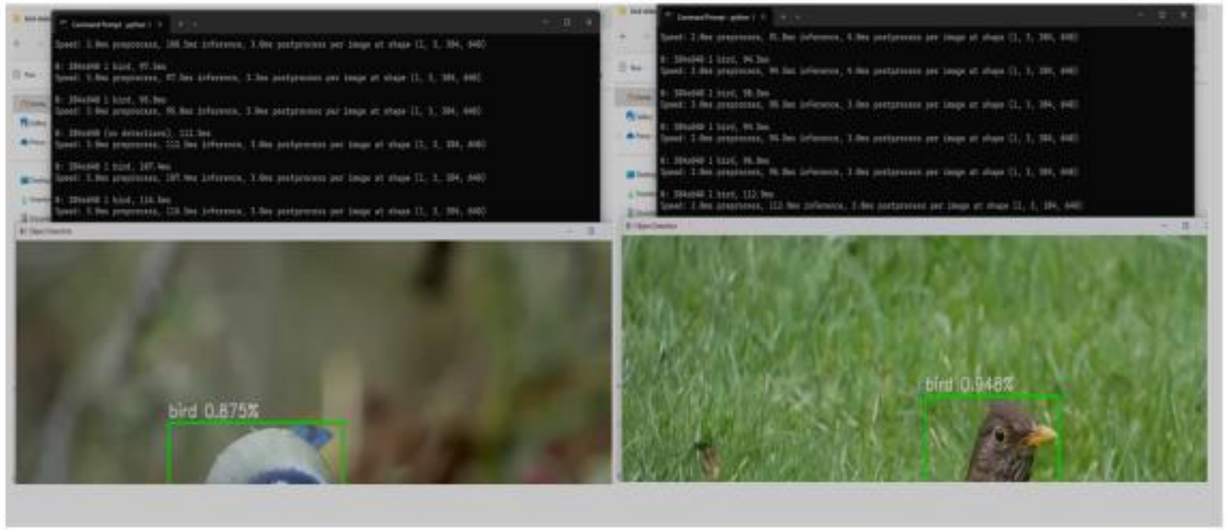
Four arm linkage

$$S+L \leq P+Q$$

Implementation of the System

Testing, Results & Analysis

With Video Frame



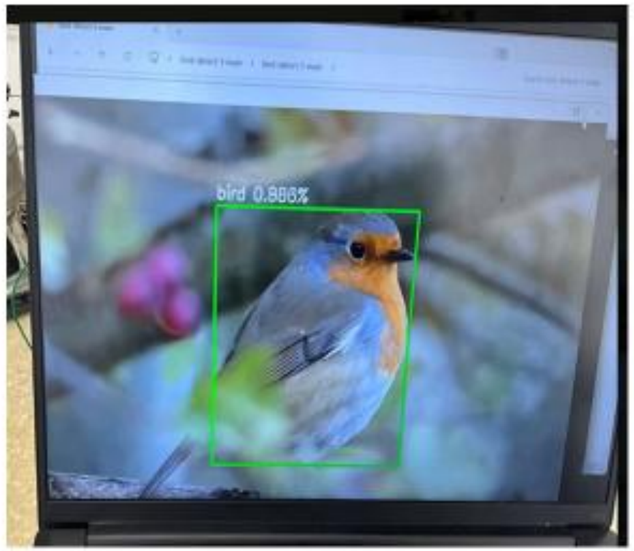
Average inference time of 3ms per frame and a confidence threshold of 0.8. Processing time: 94 ms, speed 2ms

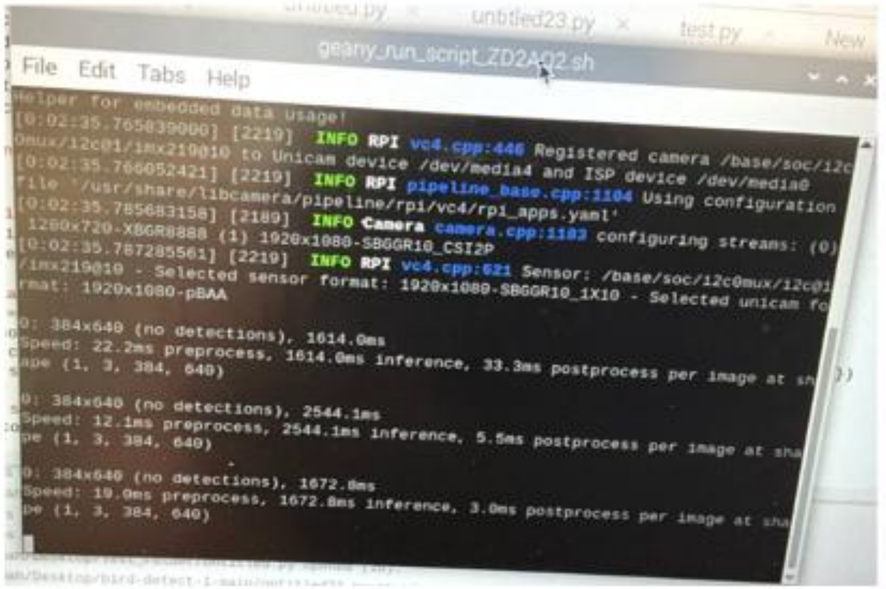
with the laptop camera

Testing, Results & Analysis

Raspberry Pi and video frames

Raspberry Pi with picamera





Average Processing time is 1.8 s & Average inference time of 13.6 milli seconds per frame

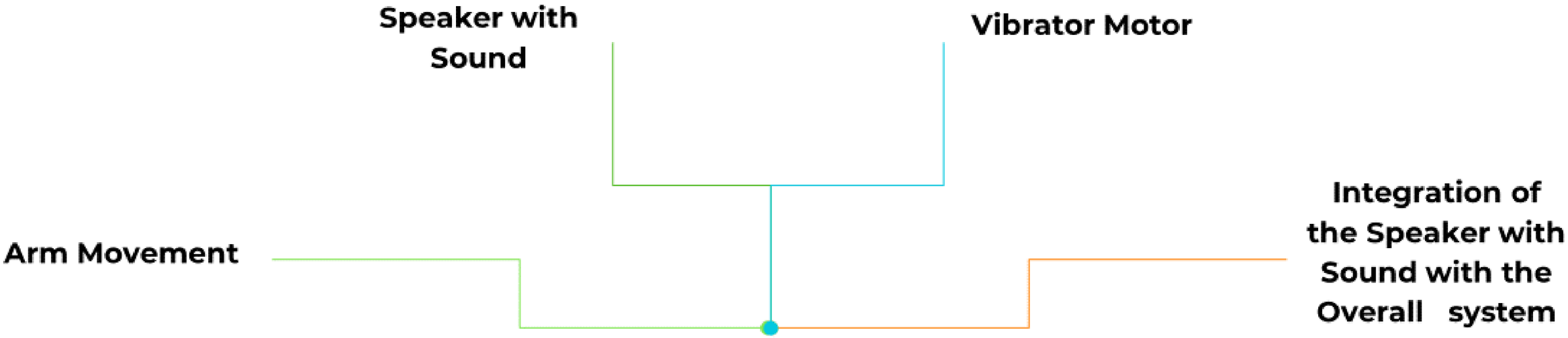
Testing, Results & Analysis

- Testing of the Vibrator Motor Was done

- Testing of the Speaker with the sound was done

- Testing of the arm movement was also done

Demonstration

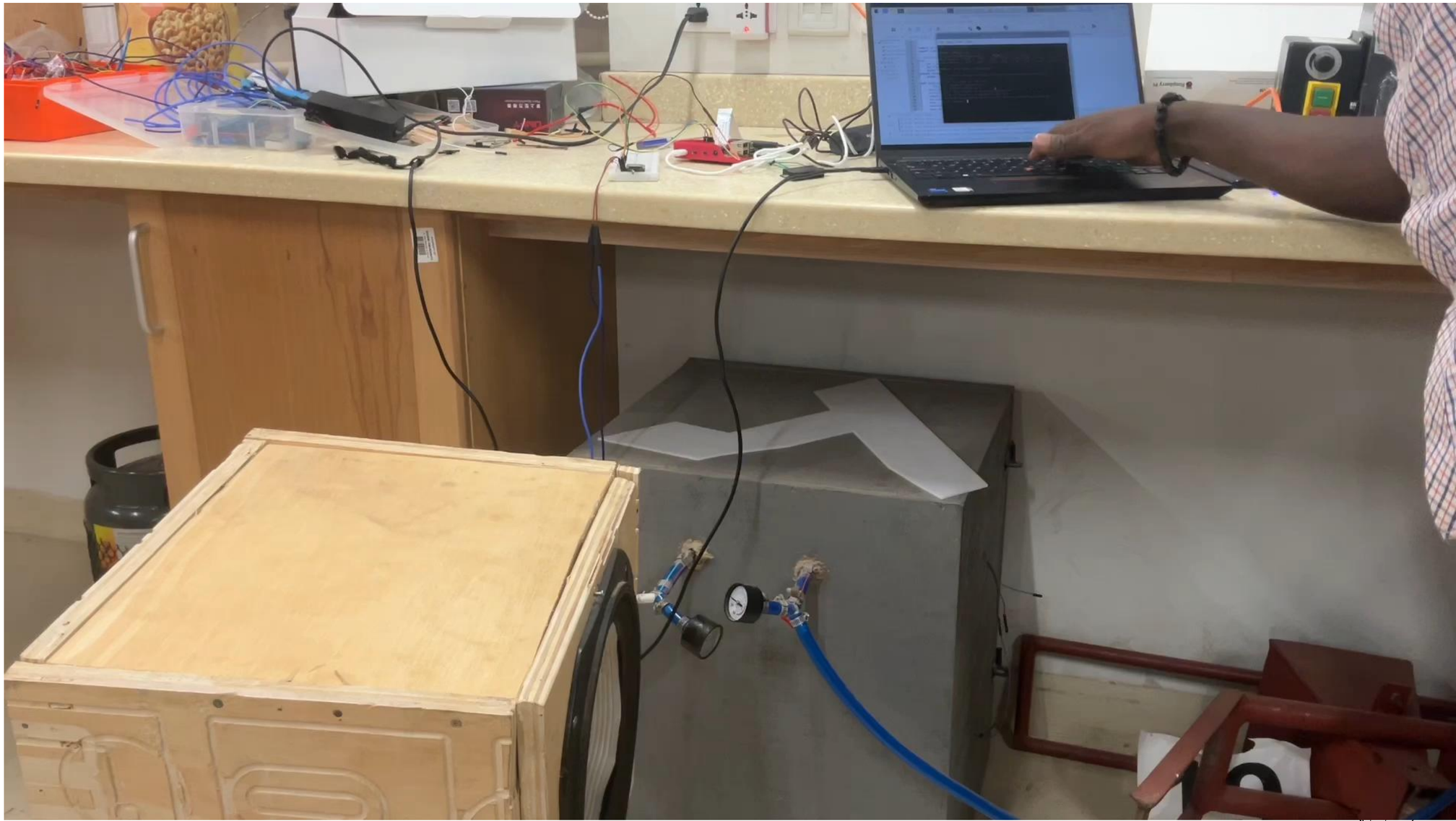


Demonstration

Arm Movement

Demonstration

Speaker with Sound



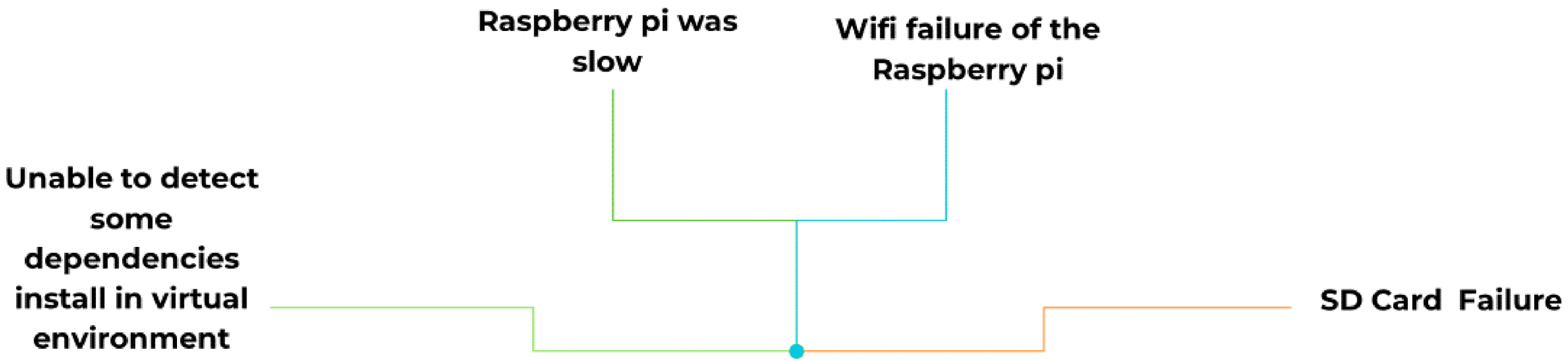
Demonstration

Vibrator Motor

Demonstration

Integration of the Speaker with Sound with the Overall system

Challenges faced



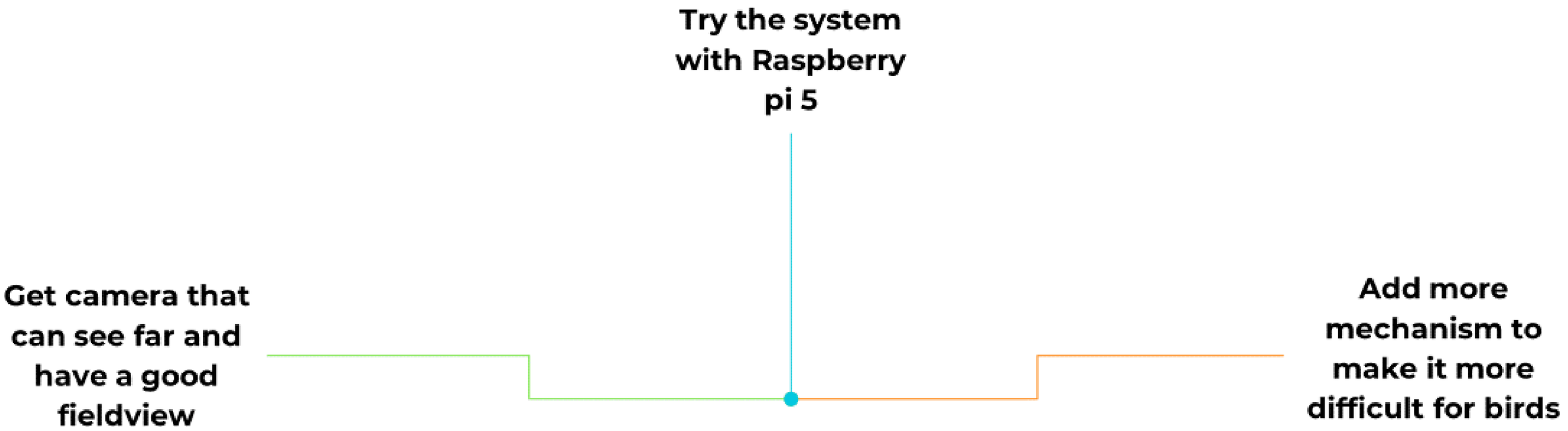
Limitations

The picamera unable to see far distance

Cloudy days limit the operation of the system

No rice farm around for actual testing

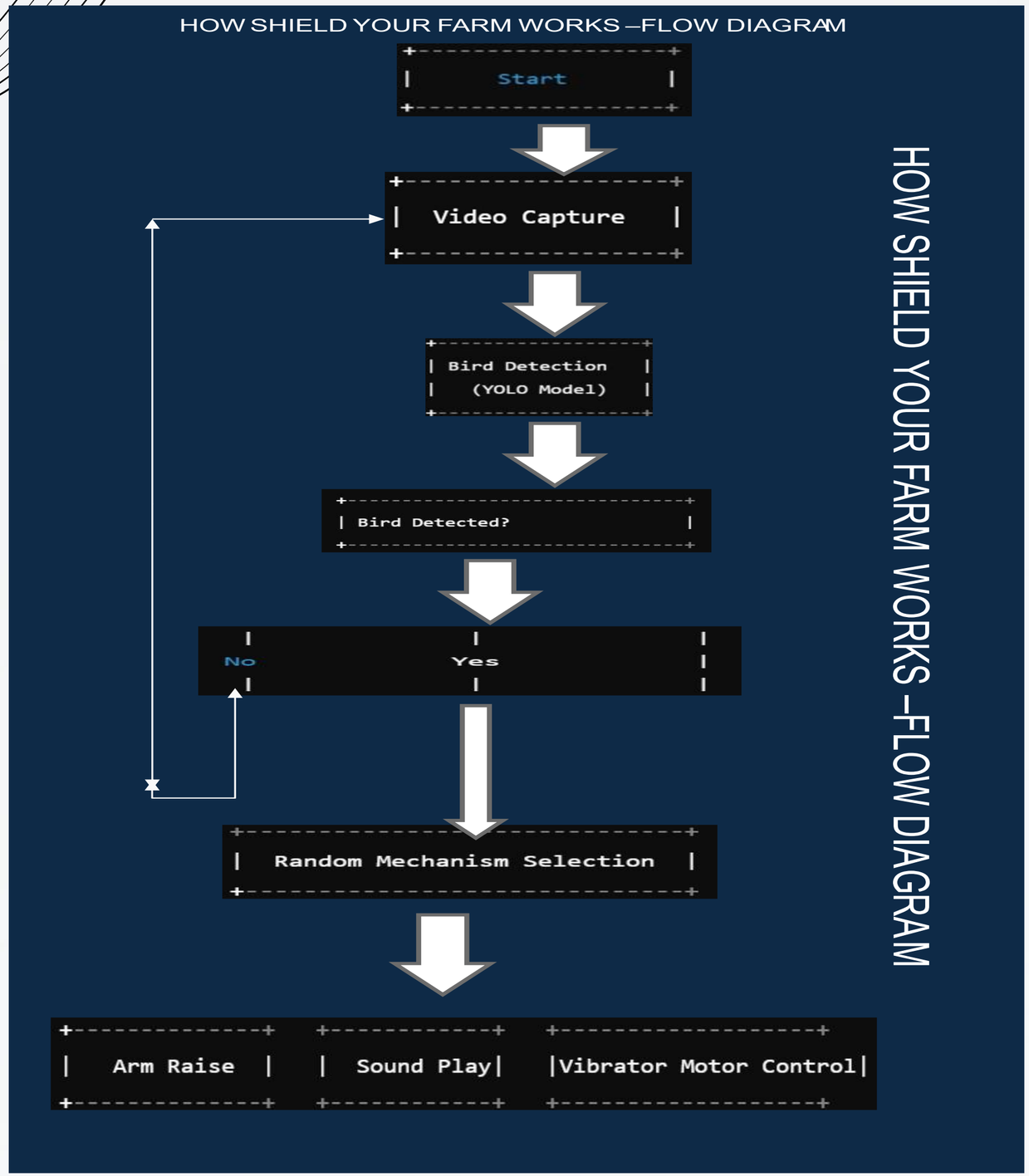
Future Works



THANK YOU!

Appendix

(255) The Human Scarecrows of Akuse (How to scare birds in rice farms) – YouTube


HOW SHIELD YOUR FARM WORKS –FLOW DIAGRAM
Start
Video Capture
Bird Detection
(YOLO Model)
Bird Detected?
No
Yes
Random Mechanism Selection
Arm Raise
Sound Play
Vibrator Motor Control
HOW SHIELD YOUR FARM WORKS –FLOW DIAGRAM

Achieved objectives

- ✓ Two dimensional arm movement mechanism
- ✓ One Traditional Approach Automation
- ✓ Using Sound Mechanism
- ✓ Leveraging on Machine Learning for bird detection

Selecting Suitable Power Source

$$PT = [14.4 + 15 + 14.4 + 0.1152 + 3]\,W$$
$$PT = 46.9152W$$

Battery Specifications:

$$Voltage = 12V$$
$$Capacity = 17Ah$$

Assumptions made

Assuming continuous operation

- Camera: (P_camera) - Servo: (P_servo)

Occasional Components:

- Sound and Arm activation - Vibrating Motor

- Total power for occasional components: (P_occ = 46.9152W -P_cont)

Power Requirement for Continuous Operation:
To run the scarecrow for 9 to 10 hours:
We calculate the continuous current draw:

$$I\_cont = P\_cont/V$$

Given the battery capacity of $17Ah$:

$$Runtime = Capacity/I\_cont$$

To ensure the scarecrow runs for 9 to 10 $hours$, we need:

$$I\_cont = 17Ah/10hours = 1.7A$$

Power Consumption of Camera and Servo:

$$Assume:$$
$$Camera: 5W$$
$$Servo: 5W$$
$$P\_cont = 5W + 5W = 10W$$

Selecting Suitable Power Source

$$I\_cont = 10W/12V = 0.833A$$

Runtime:

$$Runtime = 17/0.833 = 20.4hours$$

This is sufficient for continuous operation. So, if we account for occasional components:
Assume occasional operation for 1 hour of total active time per day:

$$Total\ occasional\ power: = 46.9152W - 10W = 36.9152W$$
$$I\_occ = 36.9152W/12V = 3.076A$$
$$Q\_occ = 3.076A * 1hour = 3.076Ah$$

Combined Runtime:

Total current draw for continuous operation over 9 $hours$:

$$I\_total = 0.833A * 9hours + 3.076Ah$$
$$= 7.497Ah + 3.076Ah$$
$$= 10.573Ah$$

The battery can support this this Scarecrow, providing:

$$Total\ runtime = 17Ah/10.573Ah/day = 1.6\ days$$

Position Analysis

- **Input Link (a)**: The link that is driven by the input motion.
- **Coupler Link (b)**: The link that connects the input and output links and often moves in a complex path.
- **Follower Link (c)**: The link that is driven by the coupler link and provides the output motion.
- **Fixed Link (d)**: The stationary frame or base of the mechanism.

- **θ1**: Angle of the input link relative to a reference frame
- **θ2**: Angle of the coupler link relative to a reference frame
- **θ3**: Angle of the follower link relative to a reference frame

Equation for Sines:

$$a\sin(\theta_1) + b\sin(\theta_2) = c\sin(\theta_3)$$

Equation for Cosines:

$$a\cos(\theta_1) + b\cos(\theta_2) = d + c\cos(\theta_3)$$

Velocity Analysis:

Differentiate the position equations with respect to time:

$$\frac{d}{dt}\left(a\cos(\theta_1) + b\cos(\theta_2)\right) = \frac{d}{dt}\left(d + c\cos(\theta_3)\right)$$

$$-a\dot{\theta_1}\sin(\theta_1) - b\dot{\theta_2}\sin(\theta_2) = -c\dot{\theta_3}\sin(\theta_3)$$

where $\dot{\theta}_i$ represents the angular velocity of link **i**.

Acceleration Analysis:

Differentiate the velocity equations with respect to time:

$$\frac{d}{dt}\left(-a\dot{\theta_1}\sin(\theta_1) - b\dot{\theta_2}\sin(\theta_2)\right) = \frac{d}{dt}\left(-c\dot{\theta_3}\sin(\theta_3)\right)$$

$$-a(\ddot{\theta_1}\sin(\theta_1) + \dot{\theta_1}^2\cos(\theta_1)) - b(\ddot{\theta_2}\sin(\theta_2) + \dot{\theta_2}^2\cos(\theta_2)) = -c(\ddot{\theta_3}\sin(\theta_3) + \dot{\theta_3}^2\cos(\theta_3))$$

where $\ddot{\theta}_i$ represents the angular acceleration of link **i**.